EECS 473
Advanced Embedded Systems

Smart Handle

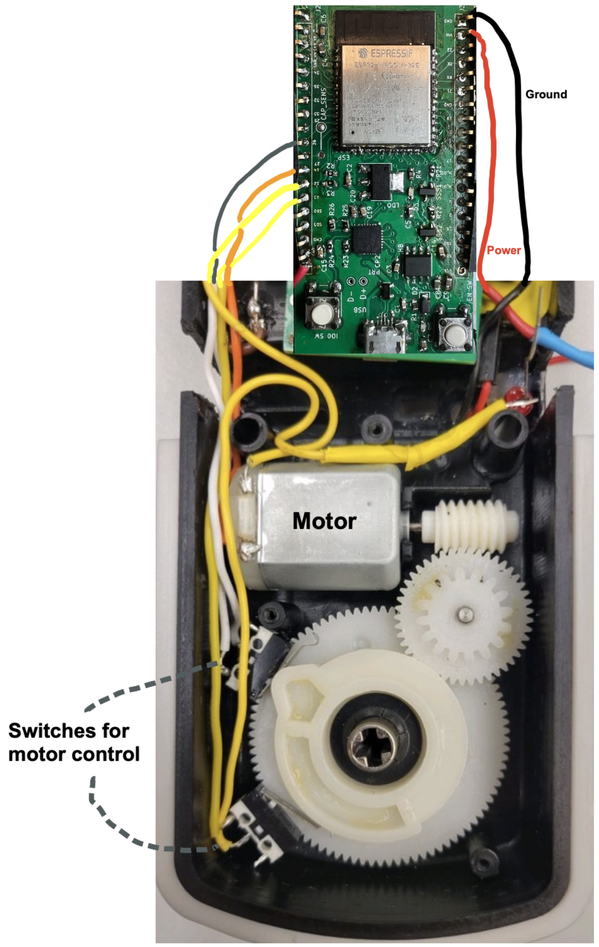
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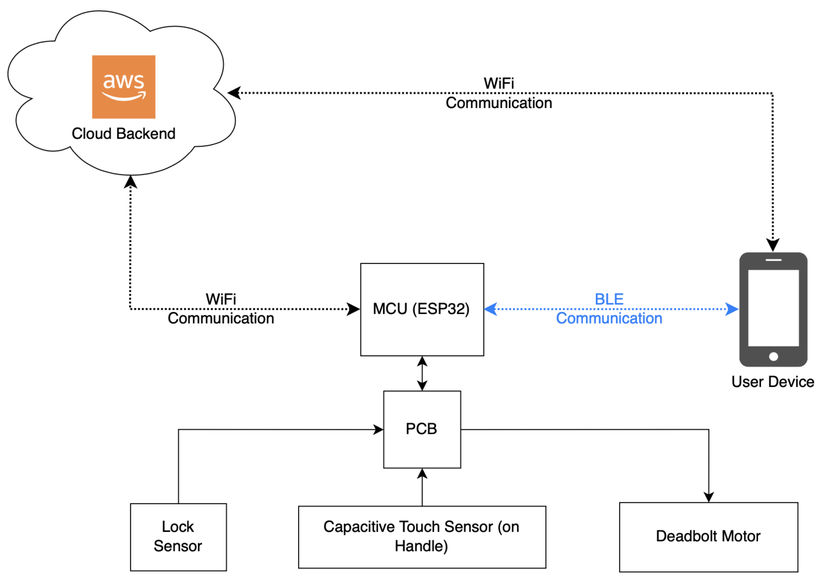
Project Objective

- Smart Handle is a smart door handle intended to simplify access control through exterior entryways
- It provides the user options on modes of entry which vary in security and convenience

Software

- **Mobile Application**
  - Application written in React Native
  - Connects to MCU via BLE and sends updated user info to the cloud
  - Allows background BLE scanning
  - Implements geofencing for proximity mode and buttons for secure mode
- **Cloud Backend**
  - AWS IoT Core for MCU - cloud communication and state persistence
  - AWS API Gateway + Lambda for Mobile - cloud communication
  - AWS Dynamodb for user information persistence
- **MCU Interfacing**
  - NimBLE, Hornbill AWS libraries used for wireless communication
  - Capacitive touch triggered ISR
  - FreeRTOS





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Thanks to Alan Tondryk for the help on the PCB
Thanks to Todd Cazallis for the help constructing the door frame

Modes of Entry

- **Proximity Mode**
  - Door unlocks automatically after touching the door
  - MCU automatically verifies identification
  - Least secure but most convenient method
- **Secure Mode**
  - User can use an app to unlock or lock the door from any location
  - Most secure but least convenient method

Hardware

- **Capacitive touch sensor**
  - Detects touch by estimating the capacitance through a GPIO pin
- **Motor & switches**
  - Motor controls lock position
  - Detects manual changes in lock position through switches positioned by the motor
  - Updates current state to cloud via WiFi
- **PCB**
  - MCU (WIFI+BLE), H-Bridge, GPIO pins

See more: https://github.com/ajsinger1/eecs473-smart-door